

Keuntae Kim

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Summary

PhD Candidate at Hanyang University working on two threads. **Interpretability & Safety**: white-box use of model internals — attention, entropy, confidence dynamics — to detect hallucination, assess reasoning quality, and steer decoding without retraining. **LLM / VLM (Video-LLM) Post-training**: RLVR and policy optimization, multimodal alignment, and continual learning, across LLMs and Video-LLMs. This has produced **3 first-author papers** at **NeurIPS 2025 / CVPR 2026 / ICML 2026** and **5+** first-author submissions under review at NeurIPS / EMNLP 2026. I also own the **distributed training stack** end-to-end, training LLMs and VLMs on **up to 32 H100s (4 nodes × 8)** across Samsung SDS Cloud, NHN Cloud (B200), and AWS (H200). **Researcher** on two Korean national AI projects (NRF, IITP); reviewer for NeurIPS, ICML, CVPR, ECCV, and ACL ARR.

Experiences

Large-Scale & Multi-Node Training Experience

Samsung SDS Cloud / NHN Cloud / AWS

Hands-on owner of end-to-end distributed training for LLMs and VLMs on three production GPU clouds — cluster bring-up, NCCL / RDMA interconnect validation, parallelism strategy, and fault-tolerant multi-day runs.

Platform	Accelerator	Topology	Scale
Samsung SDS Cloud	NVIDIA H100 SXM (80 GB)	4 nodes × 8 GPUs	32 GPUs — largest configuration; NVLink/NVSwitch intra-node, RDMA inter-node
NHN Cloud	NVIDIA B200 (Blackwell)	2 nodes × 8 GPUs	16 GPUs — next-gen Blackwell bring-up, FP8/BF16 mixed-precision training
AWS	NVIDIA H200 SXM (141 GB)	2 nodes × 8 GPUs	16 GPUs — EFA/RDMA multi-node fabric, HBM3e-enabled long-context training

- **Parallelism & memory.** Trained models beyond single-node memory with DeepSpeed ZeRO-3, PyTorch FSDP, and Megatron tensor / pipeline parallelism, combined with activation checkpointing, Liger kernels, `torch.compile`, and FlashAttention-2/3 under BF16 / FP8 mixed precision.
- **Workloads.** Full-parameter and LoRA fine-tuning of **7B–30B** LLMs via **ms-swift** and **LLaMA-Factory**; VLM / Video-LLM multimodal training (vision encoder + projector + LLM stages); and **RLVR** / **policy optimization** (GRPO, PPO) on **verl**, **NeMo-RL**, and **slime** with vLLM-backed distributed rollouts.
- **Cluster stack.** SLURM multi-node scheduling with **Ray** orchestration; **Data-Juicer** and **NeMo Curator** for large-scale data curation; **lm-evaluation-harness**, **OpenCompass**, and **EvalScope** for evaluation; **Megatron-Bridge** / **mbridge** and **checkpoint-engine** for checkpoint and weight conversion across training–inference formats; **W&B** for experiment tracking.

SP-GRPO — Signal-Preserving Aggregation for Multi-Reward RL

Work in progress

- Identified a **structural failure in multi-reward GRPO**: consensus (AND) aggregation collapses exponentially in the criterion count M — only **0.9%** of responses carried usable signal at $M=10$ — while linear averaging permits compensation and cannot express conjunctive requirements.
- Proposed a **temperature-tempered geometric mean of rank quantiles** (soft rank-product) that preserves AND semantics without signal collapse, and is scale-invariant and outlier-bounded; proved collapse and non-vanishing conditions, and that group-shared reward components vanish identically under group-relative advantage.
- On GSM8K with **8 rule-based criteria** (verl full fine-tuning, **2 nodes × 16 GPUs**), swapping only the aggregator: **0.785** vs. 0.758 (linear) and 0.699 (consensus, *below* the 0.721 base).

Implicit Attribute Installation — Mitigating Silent Attribute Loss in RL

Work in progress

- Showed that quality-only RL **silently destroys unspecified behaviors**: dialogue-containing generations fell from **19% to 0%** across 3 seeds, with lexical diversity collapsing **0.65 → 0.51**.
- Naming the attribute in the rubric invites **Goodhart** (surface mimicry scores as well as substance), so instead install it as a **latent preference in the judge**, and use **surface-vs-substance separation** as a mimicry detector.
- Established a **judge-capacity flip** (implicit channel wins precisely where judges are too weak to penalize mimicry)

— the common deployment regime), a **threshold–saturation dose-response** curve over channel strength, and a **hybrid channel** that is superadditive over either alone (+2.0 substance vs. −1.2 for explicit rubrics at 27B).

Hippocampal Memory & Sleep Consolidation for Continual Learning

Work in progress

- Built a class-incremental learner on the **complementary learning systems** account of memory: a **hippocampus-like associative matrix** binds new classes in one shot on a **frozen coordinate frame** of invariant addresses, while the backbone plays the slow-learning **neocortex** — with **no old samples, no replay buffer, and no teacher model**, and a single fixed-size memory as the only persistent state.
- Introduced **sleep consolidation**: at each task boundary the system dreams its own past — pseudo-inputs are back-generated from the memory’s principal directions and distilled into the backbone, transferring hippocampal traces into cortical weights. This was the *only* channel found to genuinely **restore** forgotten ability (up to +17.3pp on the oldest task), and, as in biology, the more fragile the learner the more it depends on sleep.
- Across **3 backbone families** (Llama-3.2-1B, Qwen3-0.6B, Gemma-3-1B) and **4 benchmarks** (TACRED, FewRel, CLINC150, Banking77) under one shared configuration: **+29 to +56 AA** over sequential fine-tuning, **exceeding the joint full-fine-tuning upper bound** on TACRED (78.7 vs. 74.1), with order-robustness and bit-level reproducibility.

CLAWS — Creativity Detection for LLM-Generated Solutions NeurIPS 2025 github.com/kkt94/CLAWS

- Proposed **CLAWS**, a white-box detector that uses attention weights across prompt sections vs. output to classify LLM solutions as *Typical* / *Creative* / *Hallucinated* — the first three-way creativity assessment for reasoning, with no human evaluation in the loop.
- Validated on **5 7–8B math RL models** over **4,545 problems from 181 contests** (AJHSME, AMC, AIME), outperforming **5 white-box baselines** (Perplexity, Logit/Window Entropy, Hidden Score, Attention Score) on both creativity and hallucination detection.
- Ensured evaluation rigor by scoring every signal under **5 evaluation strategies** (threshold, prototype, MLP, XGBoost, TabM) and reporting macro-averaged metrics to correct for the severe imbalance of the Creative and Hallucinated minority classes.

Thinking Diffusion — Visual-Grounded Reasoning for Diffusion MLLMs

CVPR 2026

- Conducted the **first quantitative analysis of CoT reasoning in dMLLMs**, identifying two failure modes: *Early Answer Generation* (model commits to the answer before reasoning) and *weak early-step visual grounding*.
- Proposed two training-free inference-time methods — **Position & Step Penalty (PSP)** and **Visual Reasoning Guidance (VRG)** — that delay premature answers and amplify visual conditioning via classifier-free guidance.
- Across LaViDa and MMaDA on *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench*: **up to +7.5%** accuracy and **>3× faster** inference vs. baselines using 4× more diffusion steps. Plug-and-play across dMLLMs with no retraining.

ST-Veto — Spatio-Temporal Token Veto for Diffusion MLLMs

ICML 2026

- Designed a training-free, decoding-time **veto-and-swap mechanism** that exploits a property unique to dMLLMs: visibility of all tokens across diffusion steps.
- Vetoes (i) *temporally unstable tokens* via second-order Taylor prediction of confidence dynamics and (ii) *weakly-grounded tokens* via image-attention mass, then swaps them for safer candidates before unmasking.
- On LaViDa-reason and MMaDA-MixCoT across *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench* under long generation lengths: **up to +9%** accuracy over baseline decoding and prior VLM reasoning methods, with **zero extra training or generation cost**.

CWM — White-Box Meta-Prompting for Adaptive RAG

EMNLP ARR (Mar 2026)

- Designed **CWM**, a low-cost white-box meta-prompting method that decides whether to retrieve using model-internal signals only — no external classifier, no multi-sampling.
- Achieves **state-of-the-art on 3 adaptive RAG benchmarks** across GPT-oss-20b, Qwen3-14b, and Llama3.1-8b, while also generalizing to pure reasoning tasks (CoT / Plan-and-Solve / Self-Discover regimes).
- Exposes a tunable internal signal that lets users control retrieval frequency at inference time without retraining.

Projector-JEPA — Gated DeltaNet MoE Projector for Video-LLMs

NeurIPS 2026 (under review)

- Proposed a **Mixture-of-Experts projector** (Local Expert + State Expert, both Gated DeltaNet) that augments any frozen Video-LLM's base projector to explicitly model inter-frame motion and global state transitions.
- Introduced **JEPA-style auxiliary objectives** — local correspondence + masked state reconstruction — to specialize each expert without touching the vision encoder or LLM.
- Trained only an **~80K-parameter** module for a **single epoch on 70K public samples**, yielding **up to ~7%** gains across **7 Video-LLMs** (PLM, VideoLLaMA3, InternVL3 families) on **5 benchmarks** (MVBench, TempCompass, NExT-QA, WorldBench, Perception Test).

Education

Hanyang University, Ph.D. in Computer Science

Sep. 2023 – Present

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Hanyang University, M.S. in Computer Science

Sep. 2021 – Aug. 2023

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Kookmin University, B.S. in Automobile and IT Convergence

Mar. 2015 – Aug. 2019

Seoul, Korea

Research Interests

- LLM / VLM (Video-LLM) Post-training
- Interpretability, Alignment & Safety
- Reinforcement Learning with Verifiable Rewards (RLVR); policy optimization for reasoning
- Video-LLMs, World Models, Vision-Language-Action models
- Continual / Incremental Learning (stability-plasticity)
- Diffusion Large/Multimodal Language Models (dLLMs, dMLLMs); visual-grounded reasoning

Publications (Journals & Conferences)

Bold + underline = author; * = first author.

1. **Keuntae Kim***, Yong Suk Choi. “CFP-AL: Combining Model Features and Prediction for Active Learning in Sentence Classification.” *Applied Sciences*, 2025. **PUBLISHED**
2. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections.” *NeurIPS 2025*. **PUBLISHED**
3. **Keuntae Kim***, Mingyu Kang, Yong Suk Choi. “Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models.” *CVPR 2026*. **PUBLISHED**
4. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding.” *ICML 2026*. **PUBLISHED**
5. **Keuntae Kim***, Eunhye Jung, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “Analyzing Creative Capabilities in RLVR Models: Overconfidence and High-Entropy Segments.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
6. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “Revisiting Incremental Learning: A Three-Interface Diagnosis of Stability and Plasticity.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
7. **Keuntae Kim***, Jeonghyun Kim, Dongjun Kim, HyeongSeok Kang, Yong Suk Choi. “Projector-JEPA: A Gated DeltaNet MoE Projector Bridging Video-LLMs to World Understanding.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
8. JungHyoun Kim, **Keuntae Kim**, HyeongSeok Kang, Dongjun Kim, Yong Suk Choi. “FourDelta: Fourier-Band Delta Predictor for Latent World Models.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
9. **Keuntae Kim***, Eunhye Jeong, Yong Suk Choi. “CWM: Controllable White-Box Meta-Prompting for Adaptive Retrieval-Augmented Generation and Reasoning Ability.” *EMNLP 2026 (under review)*. **UNDER REVIEW**

Research Grants & Funded Projects

- **Metacognitive AGI Framework and its Applications** — IITP, Korea. *Lead Researcher* (Jul. 2025 – Present).
- **Metacognitive Generative AI Framework for Mitigating Hallucination in LLMs** — NRF, Korea. *Lead*

Researcher (Feb. 2025 – Present).

- **Big Data-based Real-time Monitoring and Social Simulation System** — NRF, Korea. *Lead Researcher, Group 3* (Sep. 2021 – Feb. 2025).
- **High-performance Deep Learning via Cross-connection of Autoencoders and Decoders** — NRF, Korea. *Researcher* (Sep. 2021 – Feb. 2024).
- **Natural Language Data Refinement for Large-scale LM Construction** — KT Corp. industrial collaboration. *Researcher* (Sep. 2021 – Jul. 2022).

Patents (Korea)

- “Method and System for Fusion of Sparse-Dense and Global Attention for Video Action Recognition,” KR 10-2024-0132801 — *spatio-temporal attention fusion for video understanding*.
- “Device and Method for Correcting Toxic Words in Sentence,” KR 10-2024-0143175 — *LLM-based safety post-processing*.
- “Method and Device for Displaying Emotion Map Animation,” KR 10-2024-0152245 — *interactive web visualization of emotion trajectories over geo/time data*.
- “Method for Analysing Emotion Based on SNS Data,” KR 10-2023-0193546 — *end-to-end SNS crawling, NLP-based emotion analysis, and web-served dashboard*.

Technical Skills

Research areas. LLMs, VLMs, dLLMs / dMLLMs, Video-LLMs, World Models, RLVR, RAG, Continual Learning

Training. PyTorch, DeepSpeed (ZeRO-2/3), PyTorch FSDP, Megatron-LM, ms-swift, LLaMA-Factory, HF Accelerate, PEFT/LoRA, TRL, Liger kernels, bitsandbytes

RL & post-training. verl, NeMo-RL, slime, TRL — GRPO / PPO / RLVR pipelines with vLLM rollouts

Distributed & HPC. Multi-node training (up to 32×H100 / 4 nodes), SLURM, Ray, NCCL, InfiniBand & RoCE / AWS EFA (RDMA), torchrun elastic, tensor / pipeline parallelism, FlashAttention-2/3, BF16 & FP8, Megatron-Bridge / checkpoint-engine

Data & evaluation. Data-Juicer, NeMo Curator, lm-evaluation-harness, OpenCompass, EvalScope

GPU platforms. NVIDIA H100 / H200 / B200 (Blackwell), A100; Samsung SDS Cloud, NHN Cloud, AWS

Serving & inference. vLLM, SGLang, TensorRT-LLM, HF TGI, ONNX Runtime, llama.cpp

Languages. Python, C/C++, CUDA, JavaScript, TypeScript, Java, SQL

Web & Backend. Flask, FastAPI, Django, Node.js, React, REST APIs, WebSocket

Infra. Linux, Docker, Git, W&B, MLflow, TensorBoard, NVIDIA Nsight / PyTorch Profiler

Awards & Honors

- **NeurIPS 2025 Scholar Award**
- **BK21 FOUR Scholarship** — Korean government-funded research fellowship, 2021 – Present

Academic Service

Conference & Journal Reviewer

- **NeurIPS 2026** — Conference on Neural Information Processing Systems
- **ICML 2026** — International Conference on Machine Learning
- **CVPR 2026** — IEEE/CVF Conference on Computer Vision and Pattern Recognition
- **ECCV 2026** — European Conference on Computer Vision
- **ACL ARR — March 2026** — ACL Rolling Review
- **ACL ARR — January 2026** — ACL Rolling Review